

A Packing of 52 Planes in $\mathbb{R}^{15}$

Limin Ge*

October 8, 2026

Abstract

We give 52 2-dimensional subspaces of $\mathbb{R}^{15}$ whose smallest squared chordal distance is at least 1.765293. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 1.764201, obtained by deleting 1 of the 53 planes of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 52 subspaces of dimension 2 in $\mathbb{R}^{15}$ whose pairwise squared chordal distances $D(a, b) = 2 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 1.765293 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 15$, $n = 2$. The cell $N = 52$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 1.764201, obtained by deleting 1 of the 53 planes of the packing found by this search. The Rankin simplex bound for this cell is $2(15 - 2)/15 \cdot 52/51 = 1.767320$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m15n2N52.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 101 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

367769542011025624790214776941807802234194713640431570633693524920259322
05438914180738352053442

denominator:

2083333333333378354185546586721268346148126297504070998563604909760626441
60710718667333052420361

B Coordinates

52 blocks of 15 rows with 2 integers each; divide by 10^{12} . File: `grassmann/packings/m15n2N52.txt`.

29230491385 -574873540323	9205159288 126329892351	207662110357 181476869180	382184994939 443532414645
-123374664972 293685982106	-111346150545 -141178767629	191404743056 -136318788767	6307749573 263231995693
-17396536119 332321831673	-240501262309 -121464522628	240077710475 -126113351249	28801758305 210684672825
139121447707 268371293593	19789974136 -31376547198	-250300256096 -27172803037	-73519004230 -101593882888
68957576728 -1874425331	-349986068299 -118170824386	-177860318622 -120079101430	-64097407 -140124560381
183724765254 224926908819	214157647655 -71195875958	88425171171 254480843419	-170506262952 86533493202
-342085724841 -92554243467	286945978392 90151831050	-493772500999 498451000360	-96991225888 405966727995
649550495345 -242098221316	124407944187 -162233055144	-15812926328 -180578486097	-77971752803 -370392555305
213829745331 -249926679647	6481055395 43682596922	-444014198208 -163533437799	-429712709829 -54273885901
-490135596984 -143316668553	173880223427 382811918586	-139603832798 -360543556997	37292357711 -531771421749
90540550366 -244573855823	-408688289335 134934663722	34485837753 -7335671500	678306893562 -11701797087
-47661150788 -7302806078	-89453302619 125545903696	-244444850417 260841727795	55928680522 396895781819
-233208916410 -245331571884	237695103435 67632825033	-133703675239 -451351731999	-15889259816 361548796537
178352291510 266662624334	143137570398 100496848498	14444376904 -58490894177	-128106330937 -322569057230
-64786181258 -94155674269	-457172176947 592208445938	281989437408 -135192406403	-104076218056 274284923804
36842092851 -706562242460	323318982924 382556559147	429392957556 392796577553	295047087170 -168127423171
108355367234 238742257783	288930089019 200220066668	75548359914 166997704494	338220562586 143768126668
-215603711621 -74813372538	23738954899 -55037022018	310326109407 -9139911868	43464689479 -104180706231
219946311900 -76587119800	-426457817313 -357558980276	-53264694338 251117667280	-164206058499 145916834010
490900149092 -42784263626	-348214890084 312737742711	249032980538 -77142543783	140738576288 150293564814
-335713571958 -141816075945	-23571878525 -147457919138	-144719225349 152911672125	-323400104702 231047111104
40548166681 -21568118569	119727140558 63847667216	131623742988 163601237554	360157034956 116998383372
214315043216 302022596927	45962231202 106078823678	-6542306243 -216576186096	40791632539 -39542040275
62723009647 -349256948853	-175245574388 388043101284	-39553638796 89512344300	147697995776 277940368504
278829829691 -185194091306	-45776292707 40041919724	335092695945 -59112577388	234498283152 -92088742745
-178570399672 20220951290	17705306831 -58468051123	-248618984826 -351300367518	-225985425301 -220696311268
282470314604 94172306265	358902427198 -552025731886	210549305108 621192313073	-701711375565 -34854465602
-322314265761 79994948050	-206681534164 -203755641127	-588194658522 -45614133942	271834499118 -79974882285
367109165808 210924861529	-353881536418 153122900330	-347349436881 495835609934	-262187108911 -58822007904
242417082704 -211999748674	395805649984 184069790445	141681896549 121204792946	-166506762090 -144420772769
159734654467 526711620108	-28884870240 131511806018	296870291982 -150139226334	-115719749461 -219166209617
-134874591692 366699508803	-142596079271 -123402149655	334367852137 124091561534	124156428546 -401442278544
255402611469 -469879561397	-289722076429 -338077121787	-380144679181 346906801656	80602702976 373126163623
156185042320 -59979038287	-173443185493 -408912081858	-350519425043 43561782173	-179021712534 -215699665599
285941493756 -156205246613	30291178533 71592217326	233560295987 146791155857	-227889443209 -47776453658
-146078230511 -242158215347	-384689960246 220577200240	103261189632 -626273720	-68818253968 -53026820839
284580326295 15136980409	473685803028 212819025149	212649919241 300549689079	-289378046308 100291742169
346167540132 156803346058	-70611754590 -166502978815	-338695622937 -61384849215	25683384502 -374825127309
-171937244296 -131832887911	-49316846574 -164139057249	-100262116837 -143035893870	-150878275949 600248819108
-1233105188 -12983710759	146617336776 -371505237534	-100864648188 -512373865134	189794968 -171965415845
-441226094881 -144814688752	278096365905 -309144703851	25360451247 -60695124427	145396479217 111275887060
-446444806290 304504251091	-209024002738 -115103894385	337854591815 -40383899874	-179112087618 447627595865
-318199822533 -108372715224	450056428214 654486264886	118904988501 381158448444	-3951719753 336825884279
-187628663664 -263010520799	-78162074969 78694647550	-96302024777 340817908199	-354339220351 -248526987975
52317274134 209052995305	-209990197549 -59137596695	156903402114 -443556203568	99252195738 152034876457
-363713378363 -600979182085	-184895484390 28056431129	461118310018 -15210466161	2524787011 243979029045
-119123720822 171855063022	-64282606682 233248309982	-339168112806 -59048812085	239287325017 -117994236187
35599472460 -263552689289	-392206528234 88957021851	159845491547 183582813637	-750692121814 1262622486
2575854065 13517813082	290580288894 100575778007	-161733432598 219037268235	-112568754185 -23803870585
-163896162500 214628352635	335526570483 -281952744133	-324807904416 -11518427228	26008520983 -257879074392
447949057445 -90468037319	-164964863765 -249157933974	13101163881 279982755472	-270456103791 -25624041833
283735073144 92065605668	415569474242 -2351899727803	218317915873 -151623817115	-226863518271 -286885393603
30890779927 -227248169388	-114959421939 134418654161	182658610435 265943725475	174770914939 -553059715558
-108718573280 321274466281	59549474687 -520551483193	-2822384147 10372767182	204996126085 173176363800
83881451347 254910168504	-183380377220 302008761497	-93823198653 381108209005	-47908579968 73979695400
-109557359905 -127589949719	-88354004402 113297879377	435040703467 -513483258310	256412686162 86928288455
467004959885 -185351897096	-150133147280 -179497235490	193032117019 -87721104048	204687873853 -93435940681
230011943948 218590229666	-277874067754 -212003067702	-197201576464 -78479467299	-455006955751 -294814958745
367283878194 -188005245203	-291326164925 221329479444	308446581581 -163151621859	62689634215 -5171307746
-109746353280 -304239431170	-83208512173 13328466439	488721958778 428097563980	-143377014982 627714015160
32455630074 300727442161	-19808018181 66441269944	195827813323 331418596597	224839584767 91537911067
-389802170099 266768978871	-459166994484 463429827592	665013589953 -141389050823	326678785973 -478250005583
237600232245 -265893908544	-361625991849 -439965511769	-36253315332 -216297963417	329397694534 -3453215446
-299427422973 559810628391	-233322989619 -82860707889	273384936863 68903548000	-51021030267 161419947981
462802361302 418163451268	170795113735 186001175428	217592805433 -336648746746	301114803045 -257780610699
266905400889 -71139776825	499326111894 106650543041	-188269989380 356075416026	487028603284 388181570981
173746351322 -411435717878	81209287557 11387757339	-131753718965 -188586561543	138241005701 -89644307199

-286393126173 145479402546	-751224103289 -147350505784	-428511420650 310328176691	-59835949043 152225017308
-149787884564 32125994966	229778761530 -194422438371	113863981111 269080386414	-183477587250 56270603475
-419015223219 -10774302117	-51549163210 508527496722	39735969546 -456450710086	258437537197 261622222691
-262091745351 413538742755	-49151278462 -181862656907	381003218502 -5014044640	116359390713 337723594189
40770812796 -77472923617	-9685247417 -210208773230	-200793673073 408397992024	-66246928675 -489305160430
50513957294 -83934434559	-251715011132 -31515075743	269761856106 300069957743	352584581749 127199822847
-415036950021 325937010407	248448144362 130431812341	-132357343138 219844882715	610258227974 -119398012372
-60820537939 29652461506	-293031783375 -288932289638	65085966951 -173806660611	-230779677694 60655880581
-349516003692 -321439563420	67299108385 413940852666	10460678138 101175653915	278695252757 58927878883
-216689873534 -279658425087	-435399923922 277065990468	467553099930 -405447119189	-347806720577 -145739354011
-228250603815 -78890776301	137284795953 -295727464795	-145779093274 11943222735	-126006043030 516360417843
-464633456860 -343270896113	-280826576040 -299418687387	-242149287657 278302087865	283367470416 121877738748
73251866681 -421736163137	-21271832296 -355489216081	32693531661 -492216917265	-245002259040 142483207804
161029640108 -7846913986	-471759999787 124082930060	-203211417332 -201736256170	-1945675446 58121669886
-87563919503 70869449842	451444563243 29052984626	213723478196 132551022351	-237269715380 -357211762074
-125062098200 242084738781	240081286186 -353786229293	119590172284 107272428101	564197094657 -109028232538
470296796734 216485095016	24642444738 51391250504	-225203404528 -422409345141	-131363042885 -78137159040
-80970483085 287644648048	-98307648397 -354419038401	-104683170294 -337662141657	242249719354 -483305564564
-20684975569 315306142462	249483728656 47968931576	-348903436535 54620801873	-356550254974 9051910387
-38796101605 137310383318	57844952226 -244340442275	215655203836 295416048625	-83996366727 -351398370555
-100923164292 -355103541962	-520252842898 -127535239381	578452061281 2553184911	-132296950091 -45931786897
-75866251887 -132158516655	-31447504584 -73528578685	-174053684272 -151044845278	-19156650901 -489848104516
163270452192 43968128508	115181723632 -72068406965	-242897602649 -15468010803	346912391168 -133135892413
138922037188 -102219118915	30195751100 12184710693	90765920701 -424992321130	127654053979 -104848420168
162245026689 -410912252102	296393080579 457062346083	214725089340 -211660430471	-172586138581 -422751658758
197507685836 -213512205717	-294609785048 140832649526	-13587312229 -24999762248	301054571479 96063377341